IP2b Testing Workbook

Platform reference and supporting records

Document scope

This workbook separates desktop task records, the Meta Quest protocol and supplementary speech-input. It provides the material needed to trace statements in the evaluation to the appropriate appendix.

Platform and record map

Part	Platform and content	Appropriate use
A	Desktop scene; keyboard and mouse; task traces, ratings and interview	Analyse the desktop. Do not present the controls, times or discomfort scores as headset observations.
B	Meta Quest with two tracked controllers; task and control reference from the testing plan.	Describe the headset procedure. Use separate Quest session records for claims about actual headset performance.
C	Stationary desktop tea-table; microphone input available.	Discuss waiting, recognition errors and readability as design issues. Do not calculate actual system accuracy.

Connecting the workbook to the evaluation

Use labels such as "Appendix A, P02" for desktop and "Appendix C, P02" for speech. The repeated identifiers do not establish that the two appendices describe the same person or a single session.

Keep platform-specific conclusions separate. A Q-key mistake and an Enter-key hesitation belong to the desktop adaptation. They may suggest questions for controller interaction, but they are not observed Meta Quest input errors.

Communication boundary

In Appendix A, voice was unattempted, full Task 3 was technically blocked, and Q14 remains N/A. Appendix C adds the separate speech material referenced in the evaluation; it does not replace those records or complete the original task.

Q15 combines instructions, resource displays and chat readability. Its mean is a combined readability score, not a dedicated chat-text score.

Appendix A Desktop Case Records

Scope of the voice task

The saved scene has no assigned speech configuration, and the project review found no named configuration resource. Desktop speech still needs a runtime check. Each case therefore covers typing only. The voice step is marked N (unattempted), and the combined communication task is marked X (technically blocked).

ID	Desktop use	T1 / T2 / full T3	Seconds
P01	Frequent	I / I / X	66 / 108 / 50
P02	Occasional	A / A / X	88 / 116 / 85
P03	Occasional	I / S / X	74 / 34 / 55
P04	Frequent	A / I / X	85 / 106 / 62
P05	Frequent	I / I / X	79 / 112 / 65

Coding and timing

I = independent completion. A = completion with H1 or H2 help. P = partial completion or timeout. F = unsuccessful attempt. X = technical block. S = stopped for comfort. N = unattempted. Read the outcome and assistance columns together: a step can be incomplete even when no prompts were given.

The help column counts facilitator prompts in the order H0/H1/H2. After 10 seconds without purposeful progress, H0 repeats the goal. After a further 10 seconds or an explicit request, H1 points to a visible cue. After another 10 seconds, H2 gives the exact control. Independent completion requires every necessary step within the time limit and no H1 or H2. Record 0/0/0 for a confirmed absence of prompts, N/A for an unattempted step, and unknown for missing evidence.

Desktop adaptation of the testing plan

The desktop protocol retains the three task goals, time limits, 15 questionnaire items and assistance rules from the IP2b plan. Each core task has a local target of four independent completions across five cases. The control instructions below apply to the keyboard and mouse.

Action	Verified desktop input path
Movement and view	WASD and mouse; 1 first person / 3 third person when exchange is closed.
Exchange	F near ancestor; number keys select and Enter acquires. Mouse drag acquisition also exists. Record the route actually used.
Reward handling	Z drops/picks up; E consumes the edible reward; hold Q shows aim and release launches. No controller-to-face task.
Flight	Space initiates flight; continuing ascent may consume another segment after 0.8 s. Left Ctrl descends. Record take-off cost separately from later ascent costs.
Tea table	For the plan-aligned route, acquire the table reward and use Q placement, then click it. P preview / R placement is an alternative and must be labelled as a deviation.
Typed chat	Physical typing with Enter or on-screen SEND; table click or T/K/message icon opens chat. Record access and entry route.
Voice	Click microphone. First-person path auto-posts a successful result; third-person path can put text into a draft for SEND. Current configuration readiness is unverified/blocked.

Participant goals and start state

T1: approach the monument, obtain the energy reward, set it down, retrieve and use it, then aim pottery at the clear target. T2: enter the flight challenge, take off, try additional ascent, collect a flame, try another view if comfortable, and explain score and natural end-refill. T3: place/access the table and send/read “we can fly”; the original voice component remains unavailable and is recorded separately.

Each case starts with 50 candles, empty hands, closed menus and a first-person view. Leave sufficient energy capacity to observe the reward effect. Start the flight task at the agreed entry point. Start the communication task in a stationary first-person view. Record any facilitator positioning separately from the timed task.

Equipment and boundaries

For a live session, record the computer, display settings, input devices, microphone permission and build version. The comfort ratings here concern movement on a monitor. Use N/A for a question about an action the case did not reach, including voice input in all five cases and the view switch in P03.

P01 observation record

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	5	0/0/0
T1b Acquire	I	17	0/0/0
T1c Drop and pickup	I	31	0/0/0
T1d Consume	I	44	0/0/0
T1e Throw	I	66	0/0/0
T2a Start round	I	8	0/0/0
T2b Take-off and ascent	I	16	0/0/0
T2c Flame	I	27	0/0/0
T2d View switch	I	46	0/0/0
T2e Natural refill	I	108	0/0/0
T3a Table access	I	12	0/0/0

Substep	Code	Elapsed s	H0/H1/H2
T3b Typed message	I	34	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	38	0/0/0

Whole tasks: T1 I 66 s; T2 I 108 s; T3 X 50 s. Full T3 blocked; its typed substep remains separately interpretable.

P01 responses and task trace

Exact questionnaire item from plan	Answer
01. I noticed the ancestor when I approached the monument.	5
02. I could work out how to open the exchange tree.	4
03. I could obtain the reward I intended to select.	5
04. I could distinguish dropping an item from throwing it.	5
05. The throwing arc and landing circle helped me aim.	5
06. I understood how to consume the energy reward.	5
07. I understood the resource change after consuming the reward.	5
08. I understood how to take off and boost.	5
09. I could tell how much flight energy remained.	4
10. I understood how collecting flames affected my score.	5
11. I understood why my energy refilled when the round ended.	5
12. I remained oriented when switching views.	4
13. I could enter and send the typed message.	5
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	4

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: Third person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor noticed 5 s; F and number/Enter acquisition 17 s; Z drop/pickup 31 s; E consumption 44 s; Q preview/release 66 s. Pottery lands in the target area. Energy 10→13.

T2: start 8 s, energy refill to 15; take-off 13 s changes 15→14; one further ascent pulse changes 14→13 by 16 s. First flame 27 s; 3-key view switch 46 s. Three flames; 30 points. Natural timeout 98 s; explanation complete 108 s.

T3: reward-table acquisition and Q placement, then table click, complete 12 s. Physical typing starts 17 s; matching message sent with Enter 34 s and read back 38 s. One wrong key is corrected before send. Voice is not attempted because the preflight configuration is unavailable; final record closes 50 s.

Interview responses

Which moment was most confusing, and what did you expect? “When I aimed the pottery, the arc showed me roughly where it would land before I released Q. Reading the chat bubble took more effort. After I sent the message, I had to look closely to check that it showed what I had typed.”

Which single change would help you most? “I would enlarge just the text in the chat bubble. After pressing Enter, I want to check the whole sentence while keeping the table and the rest of the scene in view.”

Why did you prefer that view or input method? “I preferred third person for flight because I could see which way the character was facing before I turned. I only tried the keyboard at the table, so I can comment on reading the text but cannot compare it with speech.”

P02 observation record

Assigned order: Voice first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 11, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	4	0/0/0
T1b Acquire	A	42	1/1/1
T1c Drop and pickup	I	56	0/0/0
T1d Consume	I	70	0/0/0
T1e Throw	I	88	0/0/0
T2a Start round	I	8	0/0/0
T2b Take-off and ascent	A	43	1/1/1
T2c Flame	I	60	0/0/0
T2d View switch	I	75	0/0/0
T2e Natural refill	I	116	0/0/0
T3a Table access	I	15	0/0/0
T3b Typed message	A	59	1/1/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	67	0/0/0

Assistance log

Task / second	Level	Exact wording
T1b / 14	H0	What would you try next?
T1b / 24	H1	Look at the ancestor exchange control cue.
T1b / 34	H2	Press F near the ancestor, select the energy reward with its number key, then press Enter.
T2b / 18	H0	What would you try next?
T2b / 28	H1	Look at the take-off instruction.
T2b / 38	H2	Press Space to take off; a further ascent input can spend another segment.
T3b / 37	H0	What would you try next?
T3b / 47	H1	Look at the send instruction beside the draft.

P02 responses and task trace

Exact questionnaire item from plan	Answer
01. I noticed the ancestor when I approached the monument.	4
02. I could work out how to open the exchange tree.	2
03. I could obtain the reward I intended to select.	3
04. I could distinguish dropping an item from throwing it.	3
05. The throwing arc and landing circle helped me aim.	4
06. I understood how to consume the energy reward.	3
07. I understood the resource change after consuming the reward.	4
08. I understood how to take off and boost.	2
09. I could tell how much flight energy remained.	3
10. I understood how collecting flames affected my score.	4
11. I understood why my energy refilled when the round ended.	3
12. I remained oriented when switching views.	3
13. I could enter and send the typed message.	3
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	3

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: First person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: notices ancestor 4 s but cannot open exchange. Prompts at 14/24/34 s; reward acquired 42 s, Z drop/pickup 56 s, E consumption 70 s and Q throw 88 s. Pottery lands outside target. Energy 11→14. One wrong exchange key before assistance.

T2: start 8 s and refill 15; prompts occur while grounded. Take-off after H2; recorded 15→14, then a further ascent pulse 14→13 by 43 s. First flame 60 s; view switch 75 s. One flame, 10 points; timeout 98 s, explanation 116 s.

T3: acquired table, placed with Q and clicked it by 15 s. Physical typing starts 20 s and target draft is ready by 27 s; participant hesitates about sending. H0 37 s and H1 47 s; Enter sends at 59 s, readback 67 s. Voice-first assignment cannot run; actual order is typed only after the preflight gate. Record closes 85 s.

Interview responses

Which moment was most confusing, and what did you expect? “I saw the ancestor and knew I needed an energy reward, but standing beside it did not tell me that F opened the exchange. Later I had the whole sentence in the draft and stopped again. I was unsure whether Enter would send it or start a new line.”

Which single change would help you most? “I would put the key beside the action I can take right now. I needed “F to open exchange” beside the ancestor and “Enter to send” beside my finished draft. Those were the two places where I waited for help.”

Why did you prefer that view or input method? “First person made my flight direction easier to follow. With typing, I could see that my draft was ready, so I knew that sending was the part I was stuck on. I did not try voice and cannot say whether it would be easier.”

P03 observation record

Assigned order: Typed first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 12, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	6	0/0/0
T1b Acquire	I	22	0/0/0
T1c Drop and pickup	I	38	0/0/0
T1d Consume	I	56	0/0/0
T1e Throw	I	74	0/0/0
T2a Start round	I	7	0/0/0
T2b Take-off and ascent	I	16	0/0/0
T2c Flame	I	24	0/0/0
T2d View switch	N	—	N/A
T2e Natural refill	N	—	N/A
T3a Table access	I	14	0/0/0
T3b Typed message	I	39	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	44	0/0/0

P03 responses and task trace

Exact questionnaire item from plan	Answer
01. I noticed the ancestor when I approached the monument.	4
02. I could work out how to open the exchange tree.	4
03. I could obtain the reward I intended to select.	4
04. I could distinguish dropping an item from throwing it.	4
05. The throwing arc and landing circle helped me aim.	5
06. I understood how to consume the energy reward.	4
07. I understood the resource change after consuming the reward.	5
08. I understood how to take off and boost.	4
09. I could tell how much flight energy remained.	3
10. I understood how collecting flames affected my score.	4
11. I understood why my energy refilled when the round ended.	N/A
12. I remained oriented when switching views.	N/A
13. I could enter and send the typed message.	5
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	4

Chronological task trace

T1: ancestor 6 s, acquisition 22 s, drop/pickup 38 s, E consumption 56 s, Q throw 74 s. Pottery lands in target. Energy 12→15. No help.

T2: start 7 s/refill 15; take-off 12 s changes 15→14, further ascent pulse 14→13 by 16 s; first flame 24 s. At 34 s participant requests a stop during forward screen motion. No view switch or natural refill observed. One flame, 10 points.

T3: after a five-minute break, the case specifies symptoms eased to 1/10 and voluntary continuation with a stationary screen task. Table access 14 s; typing starts 20 s, send 39 s, readback 44 s. Voice not attempted due to preflight configuration gate. Record closes 55 s.

Interview responses

Which moment was most confusing, and what did you expect? “I had collected one flame and understood the score change. When the scene kept moving forwards, I began to feel uncomfortable. I wanted the movement to stop before trying another view or waiting for the round to end.”

Which single change would help you most? “I would keep a pause control visible throughout flight. Once I felt uncomfortable, I wanted to stop immediately without watching the scene move while searching for a key. A gentler camera movement would also help.”

Why did you prefer that view or input method? “I stopped before switching views, so I cannot pick a more comfortable view. After the break, the stationary scene at the table let me focus on typing the message and reading its bubble.”

P04 observation record

Assigned order: Voice first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 9, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	4	0/0/0
T1b Acquire	I	8	0/0/0
T1c Drop and pickup	A	50	1/1/0
T1d Consume	I	65	0/0/0
T1e Throw	I	85	0/0/0
T2a Start round	I	6	0/0/0
T2b Take-off and ascent	I	15	0/0/0
T2c Flame	I	22	0/0/0
T2d View switch	I	40	0/0/0
T2e Natural refill	I	106	0/0/0
T3a Table access	I	12	0/0/0
T3b Typed message	I	36	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	42	0/0/0

P04 responses and task trace

Exact questionnaire item from plan	Answer
01. I noticed the ancestor when I approached the monument.	5
02. I could work out how to open the exchange tree.	5
03. I could obtain the reward I intended to select.	5
04. I could distinguish dropping an item from throwing it.	2
05. The throwing arc and landing circle helped me aim.	4
06. I understood how to consume the energy reward.	4
07. I understood the resource change after consuming the reward.	5
08. I understood how to take off and boost.	5
09. I could tell how much flight energy remained.	5
10. I understood how collecting flames affected my score.	5
11. I understood why my energy refilled when the round ended.	5
12. I remained oriented when switching views.	4
13. I could enter and send the typed message.	5
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	4

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: Third person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor 4 s; reward acquired 8 s. Participant uses Q instead of Z and unintentionally launches reward at 16 s; no progress from 18 s. H0 28 s, H1 38 s. Recovery and deliberate Z drop/pickup 50 s, E consumption 65 s, pottery throw 85 s. Pottery lands in target. Energy 9→12.

T2: start 6 s/refill 15; take-off 11 s changes 15→14 and further ascent pulse 14→13 by 15 s. First flame 22 s, view switch 40 s. Four flames, 40 points; timeout 96 s and explanation 106 s. Brief bright collection effect described as distracting.

T3: planned voice-first; actual typed-only after voice gate. Table access 12 s, typing starts 18 s, sends 36 s, readback 42 s. While typing the participant reports a small viewpoint/movement shift; no causal diagnosis is claimed. Voice not attempted. Record closes 62 s.

Interview responses

Which moment was most confusing, and what did you expect? “I wanted to set the reward down at my feet. I pressed Q and it flew away. After the prompt I understood that Z put it down and Q aimed and launched it. Getting the reward had been straightforward, so its sudden launch surprised me.”

Which single change would help you most? “I would show “Z to drop” and “Q to aim and throw” together while I am carrying something. I need that distinction when I decide what to do with the item. By the time the throwing arc appears, I have already pressed the wrong key.”

Why did you prefer that view or input method? “I preferred third person in flight because I could see the character’s heading. When I typed at the table the view seemed to shift a little. I could not tell why, but I wanted it to stay still while I checked the message.”

P05 observation record

Assigned order: Typed first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 10, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	5	0/0/0
T1b Acquire	I	23	0/0/0
T1c Drop and pickup	I	41	0/0/0
T1d Consume	I	60	0/0/0
T1e Throw	I	79	0/0/0
T2a Start round	I	9	0/0/0
T2b Take-off and ascent	I	18	0/0/0
T2c Flame	I	31	0/0/0
T2d View switch	I	52	0/0/0
T2e Natural refill	I	112	0/0/0
T3a Table access	I	13	0/0/0
T3b Typed message	I	42	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	49	0/0/0

P05 responses and task trace

Exact questionnaire item from plan	Answer
01. I noticed the ancestor when I approached the monument.	4
02. I could work out how to open the exchange tree.	4
03. I could obtain the reward I intended to select.	4
04. I could distinguish dropping an item from throwing it.	4
05. The throwing arc and landing circle helped me aim.	4
06. I understood how to consume the energy reward.	4
07. I understood the resource change after consuming the reward.	4
08. I understood how to take off and boost.	4
09. I could tell how much flight energy remained.	3
10. I understood how collecting flames affected my score.	4
11. I understood why my energy refilled when the round ended.	4
12. I remained oriented when switching views.	3
13. I could enter and send the typed message.	4
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	3

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: First person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor 5 s, reward 23 s, drop/pickup 41 s, E consumption 60 s, Q throw 79 s. Pottery lands in target. Energy 10→13. Brief search for consumption cue, under the assistance interval.

T2: start 9 s/refill 15; take-off 14 s changes 15→14 and further ascent pulse 14→13 by 18 s. First flame 31 s, view switch 52 s. Pauses to reorient, returns to first person. Two flames, 20 points; timeout 99 s and explanation 112 s.

T3: table access 13 s; typing starts 18 s, sends 42 s after correcting one wrong key; reads bubble at 49 s. Reports small text on the screen. Voice not attempted after configuration gate; closes 65 s.

Interview responses

Which moment was most confusing, and what did you expect? “After switching views, I paused to work out which way I had been flying. At the table I corrected a wrong key before sending. The text looked small enough that I had to look closely to check the bubble against what I had typed.”

Which single change would help you most? “If I could change one thing, I would keep a heading cue visible through the view change. A brief indication of which way the character is facing would help me continue along the same route without pausing to find my direction.”

Why did you prefer that view or input method? “I tried both views and went back to first person because I could better predict which way I would move. I only used the keyboard, so I can describe correcting and sending my text but cannot compare that experience with voice.”

Appendix B Meta Quest Protocol Reference

This appendix describes the Meta Quest procedure in the IP2b testing plan. It is a platform reference, not a set of observed session outcomes. No desktop timing, rating or interview response has been relabelled as headset data.

Platform and task structure

The headset protocol uses a Meta Quest headset and two tracked controllers. Task 1 covers ancestor rewards and item use within 90 seconds. Task 2 covers flight within 120 seconds, including a 90-second round. Task 3 covers tea-table access, typed input, spoken input and readback within 90 seconds.

Control reference

Action	Desktop route in Appendix A	Meta Quest route in the plan
Movement and view	WASD and mouse; 1 or 3 changes view.	Left thumbstick moves the player; B changes view.
Exchange	F opens the tree; number keys and Enter select and acquire. Mouse drag is an alternative.	Left-controller X opens the tree. Point with the right ray and hold the right trigger, or pull the icon outward.
Drop and retrieve	Z drops or picks up the item.	A short right-trigger press and release drops the held item; point and press to pick up.
Aim and throw	Hold Q to preview; release to launch.	Hold the right trigger away from the face to preview; release to throw.
Use an edible reward	E consumes the reward.	Bring the virtual reward near the face and hold the right trigger; do not touch the headset with a controller.
Take off and boost	Space takes off or continues ascent; Left Ctrl descends.	Right-controller A takes off or boosts. Each successful boost costs one energy segment.
Typed message	Physical typing; Enter or the on-screen SEND control.	Use the scene keyboard and select SEND with controller interaction.
Spoken message	Click the microphone; unavailable in the Appendix A	Hold the left trigger for English speech input, then check the displayed result.

Limits of comparison

The two platforms share task goals, but their controls and viewing conditions differ. Desktop screen-motion discomfort does not establish headset comfort. Physical-keyboard performance does not establish virtual-keyboard performance. A control mapping explains the procedure; it does not create a measured result.

Appendix C Speech Input

Attempt and readback summary

ID	First wait	First text	Retry wait and text	Readback	Outcome
P01	3.0 s	we can fly	No retry	Correct after looking closely	Completed on first attempt
P02	3.2 s	we can find	3.0 s; we can fly	Correct after retry	Completed after retry
P03	2.9 s	we can try	3.1 s; we cannot fly	Displayed words readable; target meaning incorrect	Not completed
P04	3.1 s	we can fly	No retry	Not reliably readable at normal position	Partially completed
P05	3.0 s	we can fight	3.2 s; we can fly	Correct after retry and close checking	Completed after retry

Speech Details

P01 Waiting and reading effort

Action trace: The first target phrase produces "we can fly" after 3.0 seconds. There is no second microphone activation. The text is read correctly after moving closer to the screen.

Issue: The pause creates uncertainty about whether the microphone captured the speech. Once text appears, the small caption requires extra effort to check. Recognition is correct, but waiting and reading still add effort.

Outcome: Completed on the first attempt. Retain the waiting and readability issues rather than treating completion as an absence of difficulty.

P02 Waiting and an incorrect final word

Action trace: The first result is "we can find" after 3.2 seconds. One retry produces "we can fly" after 3.0 seconds, followed by a correct readback.

Issue: The result must be checked before the incorrect final word can be noticed. Repeating the phrase adds another recording and processing step. Caption size is not the main issue in this case.

Outcome: Completed after retrying. Keep both outputs and both waits; a corrected final message does not erase the first error.

P03 Two outputs that do not match the target

Action trace: The first result is "we can try" after 2.9 seconds. The retry produces "we cannot fly" after 3.1 seconds. Both displayed results can be read, but neither matches "we can fly".

Issue: The second output changes a positive statement into a negative one. This case concerns recognition reliability rather than a failure to display text or a caption-reading problem.

Outcome: Not completed. Speech was attempted and output appeared, so this is neither an unattempted step nor a technical block.

Speech Details Continued

P04 Correct recognition but unreliable readback

Action trace: The correct result "we can fly" appears after 3.1 seconds. The short wait is accepted. The caption is too small to read reliably from the normal viewing position, and there is no speech retry.

Issue: The output matches the target, but checking it remains difficult. Correct recognition and readable presentation are separate requirements.

Outcome: Partially completed. Keep the reading difficulty separate from recognition accuracy.

P05 Waiting recognition and reading effort

Action trace: The first result is "we can fight" after 3.0 seconds. Looking closer reveals the incorrect word. A retry produces "we can fly" after 3.2 seconds, followed by another careful check and a correct readback.

Issue: Small captions delay error detection. Correcting the error then requires another spoken attempt and another wait. The three issues combine to add steps to a short message.

Outcome: Completed after retrying, with reading difficulty retained. This does not demonstrate reliable first-attempt recognition.

Use in the evaluation

Refer to these records as "Appendix C speech ". The design discussion can address processing feedback, editable drafts, retry controls and readable captions. Any claim about measured latency, real user preference or actual speech accuracy requires separate session evidence.

Keep Q14 in Appendix A as N/A. Do not pool the outcomes with the original desktop task totals, and do not replace the full Task 3 technical-block status. The identifiers are reused only to maintain the existing references.